

Yashvardhan Khanna

Bangalore | +91 9772292339 | yashvardhankhanna360@gmail.com
github.com/YashK3086 | linkedin.com/in/yashvardhan-khanna | credly.com/yashvardhan-khanna

EDUCATION

SRM Institute of Science and Technology

B.Tech in Computer Science (Cloud Computing) — First Class (70%+ Eligibility)

Chennai, TN

2023 – 2027

Delhi Public School

Class XII (CBSE): 83.4% — Class X (CBSE): 90.2%

Jodhpur, RJ

2021 – 2023

TECHNICAL SKILLS

Full-Stack Engineering: Next.js, React.js, Node.js, MERN Stack, JavaScript, HTML/CSS, Core Java

Data Science & AI: Python, PySpark, Librosa, SageMaker, KNN Models, Prompt Engineering, Hadoop

Cloud, DevOps & Data: AWS (EC2, Lambda, S3, VPC, Cognito), DynamoDB, SQL, Jenkins, Terraform, Docker

Tools & Automation: Git/GitHub, Linux (Bash), Prometheus, Grafana, ArgoCD, n8n SaaS Automation, Agile playoffs

TECHNICAL PROJECTS

Smart Serverless SaaS Music App with Context Filtering | *Next.js, SageMaker, AWS Glue, Lambda*

- Engineered a data engineering pipeline using AWS Glue & PySpark to execute Librosa audio signal processing (STFT/RMS) for automated feature extraction over rolling 10-second intervals.
- Deployed a SageMaker KNN machine learning model integrated with a serverless Lambda re-ranking architecture, driving real-time personalized user streaming content with a **20% discovery boost**.
- Optimized full-stack user states via a global Next.js AudioContext state machine, integrating debounced event handling and optimistic UI updates for real-time interface rendering.
- Built a low-latency context filter tracking user session matrices through DynamoDB TTL configurations, maintaining high performance under scaling metrics with sub-second backend execution.

Automated Blue-Green SaaS CI/CD Pipeline | *AWS EKS, Jenkins, Terraform, Argo Rollouts*

- Architected a scalable cloud deployment pipeline using Jenkins, GitOps, and Terraform to guarantee **100% application availability** during multi-tenant SaaS application version rollouts.
- Programmed automated live production assertions using custom Analysis Templates, dropping rollback execution windows to **under 60s** when identifying active HTTP faults.
- Integrated automated Slack alerts and system telemetry, increasing cross-functional agile issue response speed by **40%**.

IoT Security: Stream Cipher for LoRa E2EE | *ESP32, ChaCha20, Agile Methodology*

- Developed a 256-bit lightweight end-to-end encryption protocol for localized data transmission links to safely bypass standard middleware dependencies on resource-constrained microcontrollers.
- Created a custom 96-bit Dynamic Nonce synchronization pipeline, cutting baseline data overhead by eliminating **20% packet bloat**.
- Achieved an **18% device energy efficiency optimization** by shifting complex cryptographic overhead from block configurations into a stream-based deployment model.

Corporate Visitor Data Warehouse | *SQL, Data Warehousing, ETL Triggers*

- Designed a robust BCNF-normalized data warehouse schema, reducing system data redundancy by **35%** while significantly improving query processing throughput.
- Automated relational state workflows using custom ETL triggers and system database cursors to harden runtime storage access log validation logs.

CERTIFICATIONS & LEADERSHIP

Certifications: Cisco Ethical Hacker; Pursuing AWS Solutions Architect; Python, SQL, and Cloud Architecture.

Leadership: Head Boy (School Leadership); MUN Organizing Team; National-level Athlete.